Stratification with typed products

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Introduction

Many epidemiological models stratify a base compartmental model by attributes like age, risk group, or vaccination status. Doing this by hand is error-prone: an SIR model stratified by 3 age groups has 9 species and many transitions to track.

algebraicodin automates stratification using **typed Petri nets** and their **typed product**, porting the approach from AlgebraicPetri.jl. This is the categorical product in the slice category Petri/T, where T is a *type system* (ontology) that specifies the allowed transition types.

```
library(algebraicodin)
```

Type systems (ontologies)

A **type system** is a Petri net that defines abstract transition patterns. The `infectious_ontology()` provides the standard type system for infectious disease models:

```
ont <- infectious_ontology()
species_names(ont)
#> [1] "Pop"
transition_names(ont)
#> [1] "infect"  "disease" "strata"
```

The ontology has one species type (**Pop**) and three transition types:

Type	Input → Output	Meaning
infect	Pop, Pop → Pop, Pop	Frequency-dependent transmission
disease	Pop → Pop	Spontaneous transition (recovery, death, ...)
strata	Pop → Pop	Movement between strata (aging, migration, ...)

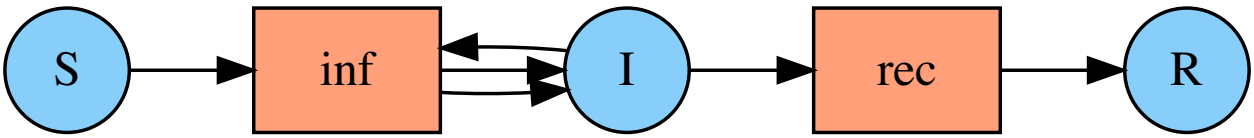
Typed Petri nets

A **typed Petri net** is a morphism $\phi : P \to T$ from a concrete Petri net to the type system. Each species and transition in P is assigned a type in T.

```
sir <- labelled_petri_net(
  c("S", "I", "R"),
  inf = c("S", "I") %=>% c("I", "I"),
  rec = "I" %=>% "R"
)
sir_typed <- typed_petri(sir, ont,
```

```
  species_types    = c(S = "Pop", I = "Pop", R = "Pop"),
  transition_types = c(inf = "infect", rec = "disease")
)
sir_typed@species_map
#> [1] 1 1 1
sir_typed@transition_map
#> [1] 1 2
```

```
plot_petri(sir_typed)
```



All SIR species map to `Pop` (index 1). The `inf` transition maps to `infect` (index 1) and `rec` maps to `disease` (index 2).

Reflexive transitions

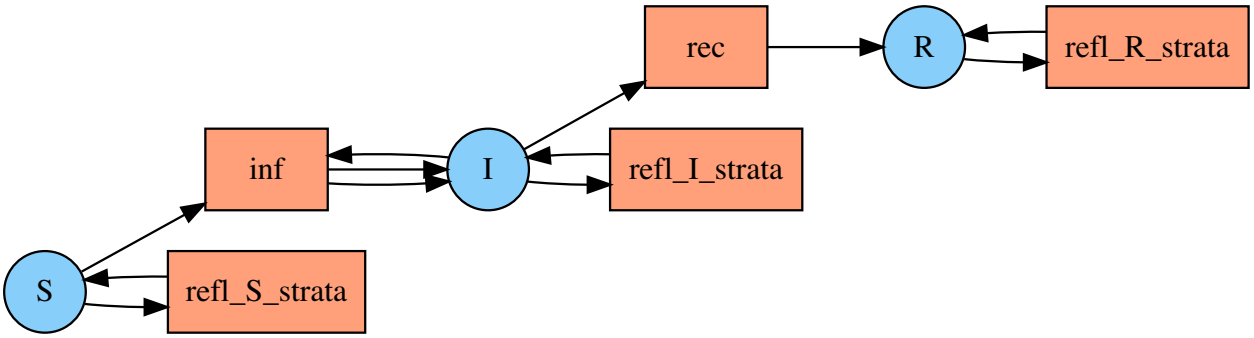
Before taking a typed product, each model must be augmented with **reflexive transitions**—identity self-loops that allow the product to pair "real" transitions in one model with "do-nothing" transitions in the other.

For stratification by age, the SIR model needs reflexive `strata` transitions (so that aging can be paired with each compartment):

```
sir_aug <- add_reflexives(sir_typed, list(
  S = "strata",
  I = "strata",
  R = "strata"
))
transition_names(sir_aug@pn)
#> [1] "inf"           "rec"           "refl_S_strata" "refl_I_strata"
#> [5] "refl_R_strata"
```

The augmented model now has 5 transitions:

```
plot_petri(sir_aug)
```



Defining strata

Age groups are modelled as a simple Petri net with `strata`-typed transitions. For two age groups with aging:

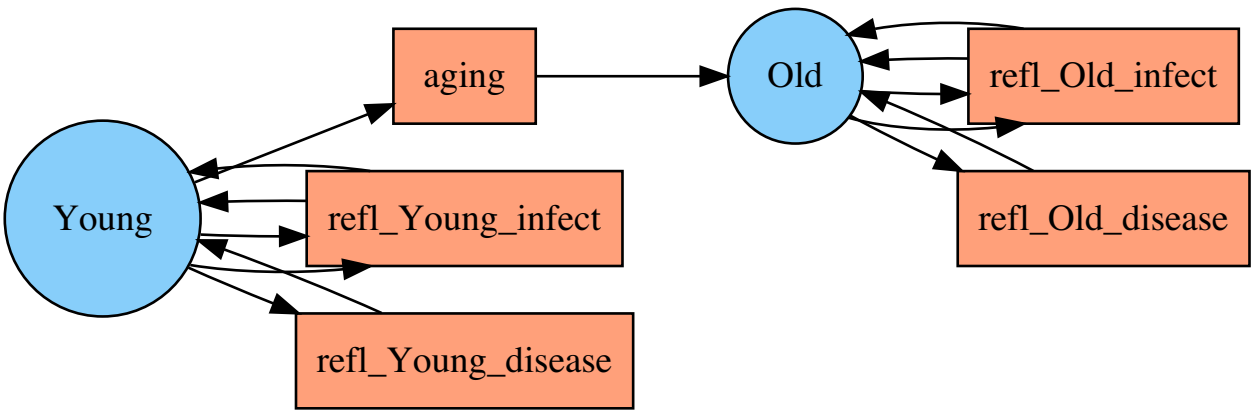
```
age <- labelled_petri_net(
  c("Young", "Old"),
  aging = "Young" %=>% "Old"
```

```r
)
age_typed <- typed_petri(age, ont,
  species_types    = c(Young = "Pop", Old = "Pop"),
  transition_types = c(aging = "strata")
)
```

The age model also needs reflexive transitions for `infect` and `disease` so these processes can occur within each age group:

```r
age_aug <- add_reflexives(age_typed, list(
  Young = c("infect", "disease"),
  Old   = c("infect", "disease")
))
transition_names(age_aug@pn)
#> [1] "aging"              "refl_Young_infect"  "refl_Young_disease"
#> [4] "refl_Old_infect"    "refl_Old_disease"
```

```r
plot_petri(age_aug)
```



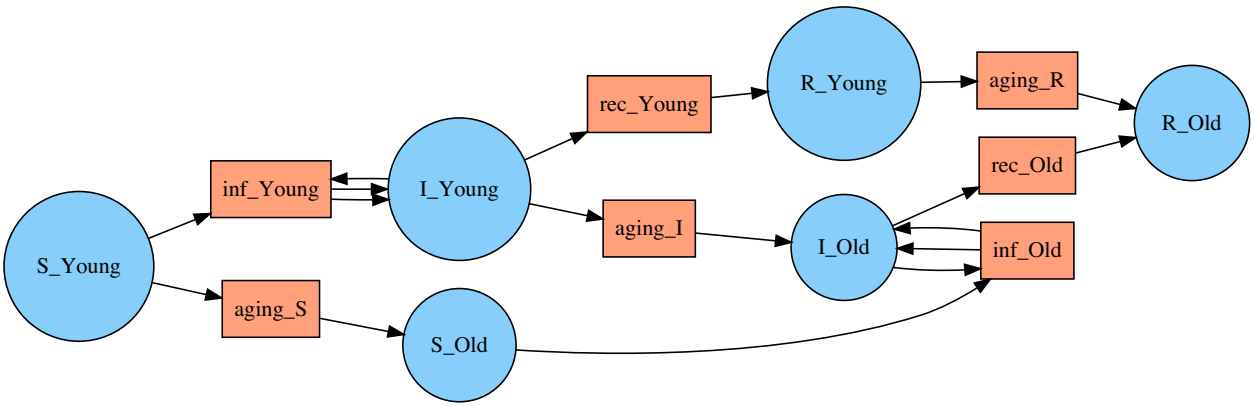
Taking the typed product

The **typed product** (or stratification) creates all valid pairs of species and transitions. The `%x%` operator is shorthand for `typed_product()`:

```r
sir_age <- sir_aug %x% age_aug
species_names(sir_age)
#> [1] "S_Young" "S_Old"   "I_Young" "I_Old"   "R_Young" "R_Old"
transition_names(sir_age)
#> [1] "inf_Young" "inf_Old"   "rec_Young" "rec_Old"   "aging_S"   "aging_I"
#> [7] "aging_R"
```

The stratified model has 6 species (S, I, R for each age group) and 7 transitions (within-group infection, cross-group infection, and recovery for each group, plus age transitions).

```r
plot_petri(sir_age)
```



The result has:

- **6 species**: S×{Young,Old}, I×{Young,Old}, R×{Young,Old}
- **7 transitions**: infection and recovery *within* each age group, plus aging *across* each compartment

Generated odin2 code

```
code <- pn_to_odin(sir_age, "ode")
cat(code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf_Young <- parameter()
#> inf_Old <- parameter()
#> rec_Young <- parameter()
#> rec_Old <- parameter()
#> aging_S <- parameter()
#> aging_I <- parameter()
#> aging_R <- parameter()
#>
#> ## Initial conditions
#> S_Young0 <- parameter()
#> S_Old0 <- parameter()
#> I_Young0 <- parameter()
#> I_Old0 <- parameter()
#> R_Young0 <- parameter()
#> R_Old0 <- parameter()
#>
#> initial(S_Young) <- S_Young0
#> initial(S_Old) <- S_Old0
#> initial(I_Young) <- I_Young0
#> initial(I_Old) <- I_Old0
#> initial(R_Young) <- R_Young0
#> initial(R_Old) <- R_Old0
#>
#> ## Transition rates
#> rate_inf_Young <- inf_Young * S_Young * I_Young
#> rate_inf_Old <- inf_Old * S_Old * I_Old
#> rate_rec_Young <- rec_Young * I_Young
#> rate_rec_Old <- rec_Old * I_Old
#> rate_aging_S <- aging_S * S_Young
```

```
#> rate_aging_I <- aging_I * I_Young
#> rate_aging_R <- aging_R * R_Young
#>
#> ## Derivatives
#> deriv(S_Young) <- -rate_inf_Young - rate_aging_S
#> deriv(S_Old) <- -rate_inf_Old + rate_aging_S
#> deriv(I_Young) <- rate_inf_Young - rate_rec_Young - rate_aging_I
#> deriv(I_Old) <- rate_inf_Old - rate_rec_Old + rate_aging_I
#> deriv(R_Young) <- rate_rec_Young - rate_aging_R
#> deriv(R_Old) <- rate_rec_Old + rate_aging_R
```

Each transition rate uses mass-action kinetics within the appropriate stratum. The aging transitions move individuals between age groups while preserving their disease state.

Comparison with hand-written code

Here is the equivalent hand-written odin2 model:

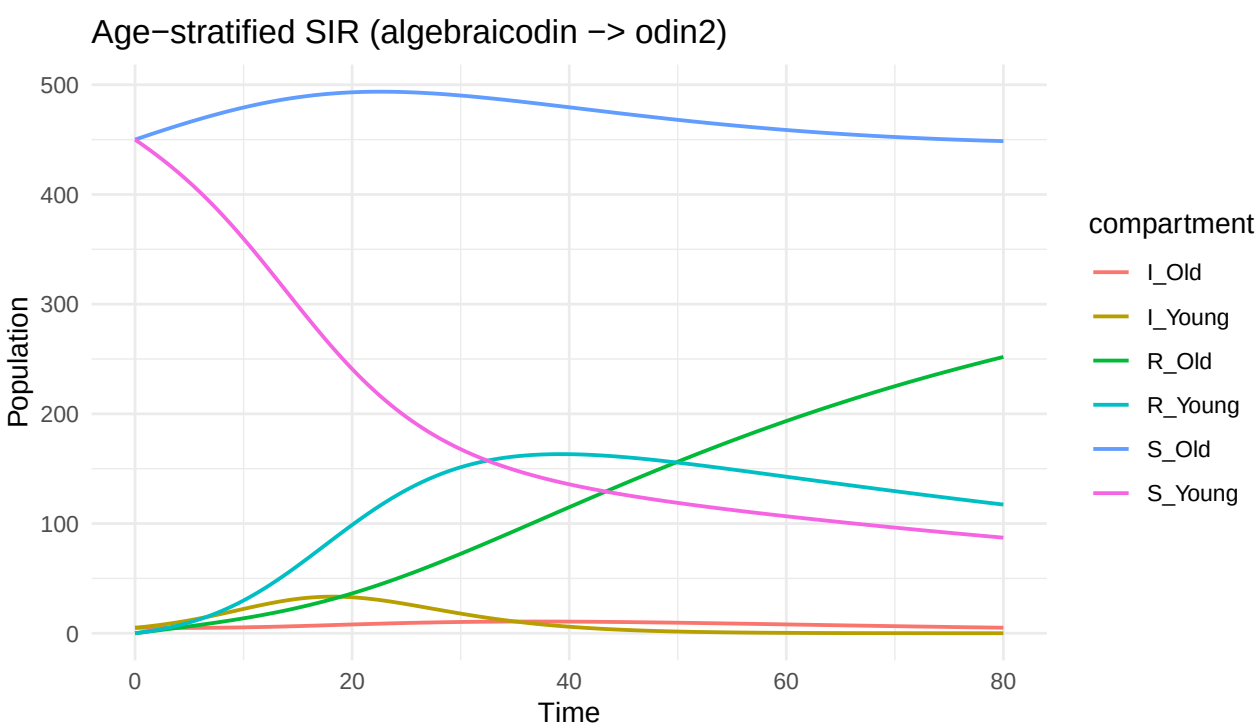
```
handwritten <- "
inf_Young <- parameter()
inf_Old <- parameter()
rec_Young <- parameter()
rec_Old <- parameter()
aging_S <- parameter()
aging_I <- parameter()
aging_R <- parameter()

S_Young0 <- parameter()
S_Old0 <- parameter()
I_Young0 <- parameter()
I_Old0 <- parameter()
R_Young0 <- parameter()
R_Old0 <- parameter()

initial(S_Young) <- S_Young0
initial(S_Old) <- S_Old0
initial(I_Young) <- I_Young0
initial(I_Old) <- I_Old0
initial(R_Young) <- R_Young0
initial(R_Old) <- R_Old0

rate_inf_Young <- inf_Young * S_Young * I_Young
rate_inf_Old <- inf_Old * S_Old * I_Old
rate_rec_Young <- rec_Young * I_Young
rate_rec_Old <- rec_Old * I_Old
rate_aging_S <- aging_S * S_Young
rate_aging_I <- aging_I * I_Young
rate_aging_R <- aging_R * R_Young

deriv(S_Young) <- -rate_inf_Young - rate_aging_S
deriv(S_Old) <- -rate_inf_Old + rate_aging_S
deriv(I_Young) <- rate_inf_Young - rate_rec_Young - rate_aging_I
deriv(I_Old) <- rate_inf_Old - rate_rec_Old + rate_aging_I
deriv(R_Young) <- rate_rec_Young - rate_aging_R
```

```
deriv(R_Old) <- rate_rec_Old + rate_aging_R
"
```

Running both:

```
pars <- list(
  inf_Young = 0.001, inf_Old = 0.0005,
  rec_Young = 0.25, rec_Old = 0.25,
  aging_S = 0.01, aging_I = 0.01, aging_R = 0.01,
  S_Young0 = 450, S_Old0 = 450,
  I_Young0 = 5, I_Old0 = 5,
  R_Young0 = 0, R_Old0 = 0
)
t <- seq(0, 80, by = 0.5)

gen_alg <- odin2::odin(code)
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system7ebba6e1
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system7ebba6e1’ ...
#>    ** this is package ‘odin.system7ebba6e1’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:128:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                           ^
#>    dust.cpp:132:20: note: in instantiation of function template specialization 'dust2::r::dust2_cont
#>    132 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system7ebba6e1) (500ms)
#>
```

```r
#> i Loading odin.system7ebba6e1
sys_alg <- dust2::dust_system_create(gen_alg, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys_alg)
y_alg <- dust2::dust_system_simulate(sys_alg, t)

gen_hw <- odin2::odin(handwritten)
#> i Using cached generator
sys_hw <- dust2::dust_system_create(gen_hw, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys_hw)
y_hw <- dust2::dust_system_simulate(sys_hw, t)

cat("Max absolute difference:", max(abs(y_alg - y_hw)), "\n")
#> Max absolute difference: 0

# Plot
snames <- species_names(sir_age)
df <- data.frame(
  time = rep(t, length(snames)),
  value = as.vector(t(y_alg)),
  compartment = rep(snames, each = length(t))
)
library(ggplot2)
ggplot(df, aes(time, value, colour = compartment)) +
  geom_line(linewidth = 0.7) +
  labs(title = "Age-stratified SIR (algebraicodin → odin2)",
       x = "Time", y = "Population") +
  theme_minimal()
```



Three age groups

Scaling to more strata is trivial. Here is SIR $\times$ 3 age groups with pairwise aging:

```r
age3 <- labelled_petri_net(
  c("Young", "Middle", "Old"),
  aging_YM = "Young" %=>% "Middle",
  aging_MO = "Middle" %=>% "Old"
)
age3_typed <- typed_petri(age3, ont,
  species_types    = c(Young = "Pop", Middle = "Pop", Old = "Pop"),
  transition_types = c(aging_YM = "strata", aging_MO = "strata")
)
age3_aug <- add_reflexives(age3_typed, list(
  Young  = c("infect", "disease"),
  Middle = c("infect", "disease"),
  Old    = c("infect", "disease")
))

sir_age3 <- sir_aug %x% age3_aug
cat("Species:", species_names(sir_age3), "\n")
#> Species: S_Young S_Middle S_Old I_Young I_Middle I_Old R_Young R_Middle R_Old
cat("Transitions:", transition_names(sir_age3), "\n")
#> Transitions: inf_Young inf_Middle inf_Old rec_Young rec_Middle rec_Old aging_YM_S aging_MO_S aging_Y
cat("nparts: S =", acsets::nparts(sir_age3@pn, "S"),
    ", T =", acsets::nparts(sir_age3@pn, "T"), "\n")
#> nparts: S = 9 , T = 12
```

9 species and 12 transitions—automatically generated with correct stoichiometry.

Risk group stratification

Risk groups follow the same pattern. Here we stratify SIR by two risk levels without movement between groups (no strata transitions):

```r
risk <- labelled_petri_net(c("High", "Low"))
risk_typed <- typed_petri(risk, ont,
  species_types    = c(High = "Pop", Low = "Pop"),
  transition_types = setNames(character(0), character(0))
)
risk_aug <- add_reflexives(risk_typed, list(
  High = c("infect", "disease"),
  Low  = c("infect", "disease")
))

sir_risk <- sir_aug %x% risk_aug
species_names(sir_risk)
#> [1] "S_High" "S_Low"  "I_High" "I_Low"  "R_High" "R_Low"
transition_names(sir_risk)
#> [1] "inf_High" "inf_Low"  "rec_High" "rec_Low"
```

Without strata transitions, there is no movement between risk groups. The infection and recovery happen independently within each group.

The `stratify()` wrapper

`stratify()` is an alias for `typed_product()`:

```r
sir_age2 <- stratify(sir_aug, age_aug)
identical(species_names(sir_age2), species_names(sir_age))
#> [1] TRUE
```

Stochastic stratified models

The stratified Petri net can generate any model type. Here is the age-stratified SIR as a stochastic model:

```r
cat(pn_to_odin(sir_age, "stochastic"))
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf_Young <- parameter()
#> inf_Old <- parameter()
#> rec_Young <- parameter()
#> rec_Old <- parameter()
#> aging_S <- parameter()
#> aging_I <- parameter()
#> aging_R <- parameter()
#>
#> ## Initial conditions
#> S_Young0 <- parameter()
#> S_Old0 <- parameter()
#> I_Young0 <- parameter()
#> I_Old0 <- parameter()
#> R_Young0 <- parameter()
#> R_Old0 <- parameter()
#>
#> initial(S_Young) <- S_Young0
#> initial(S_Old) <- S_Old0
#> initial(I_Young) <- I_Young0
#> initial(I_Old) <- I_Old0
#> initial(R_Young) <- R_Young0
#> initial(R_Old) <- R_Old0
#>
#> ## Transition probabilities and draws
#> p_inf_Young <- 1 - exp(-(inf_Young * I_Young) * dt)
#> n_inf_Young <- Binomial(S_Young, p_inf_Young)
#> p_inf_Old <- 1 - exp(-(inf_Old * I_Old) * dt)
#> n_inf_Old <- Binomial(S_Old, p_inf_Old)
#> p_rec_Young <- 1 - exp(-(rec_Young) * dt)
#> n_rec_Young <- Binomial(I_Young, p_rec_Young)
#> p_rec_Old <- 1 - exp(-(rec_Old) * dt)
#> n_rec_Old <- Binomial(I_Old, p_rec_Old)
#> p_aging_S <- 1 - exp(-(aging_S) * dt)
#> n_aging_S <- Binomial(S_Young, p_aging_S)
#> p_aging_I <- 1 - exp(-(aging_I) * dt)
#> n_aging_I <- Binomial(I_Young, p_aging_I)
#> p_aging_R <- 1 - exp(-(aging_R) * dt)
#> n_aging_R <- Binomial(R_Young, p_aging_R)
#>
#> ## Update equations
#> update(S_Young) <- S_Young - n_inf_Young - n_aging_S
#> update(S_Old) <- S_Old - n_inf_Old + n_aging_S
```

```
#> update(I_Young) <- I_Young + n_inf_Young - n_rec_Young - n_aging_I
#> update(I_Old) <- I_Old + n_inf_Old - n_rec_Old + n_aging_I
#> update(R_Young) <- R_Young + n_rec_Young - n_aging_R
#> update(R_Old) <- R_Old + n_rec_Old + n_aging_R
```

Summary

Step	Function	Purpose
Define type system	`infectious_ontology()`	Declare allowed transition types
Assign types	`typed_petri()`	Create morphism PN → ontology
Add reflexives	`add_reflexives()`	Enable cross-model pairing
Stratify	`%x%` or `typed_product()`	Compute categorical product
Flatten	`flatten()`	Extract underlying PN from typed PN
Generate code	`pn_to_odin()`	Produce odin2 DSL for any model type

The typed product guarantees structural correctness: every species pair has consistent transitions, and the stoichiometry is preserved. Scaling from 2 to 20 strata requires only changing the strata model—the base disease model is unchanged.